





Global Supply Chain Performance & Optimization Analytics

End-to-End Data Analytics Case Study

BUSINESS CONTEXT

- NexaDistrib International
- Global supply chain company 20+ Suppliers
- 8 Warehouses
- 100+ Customers
- Multi-region operations

BUSINESS PROBLEMS

Key Business Problems

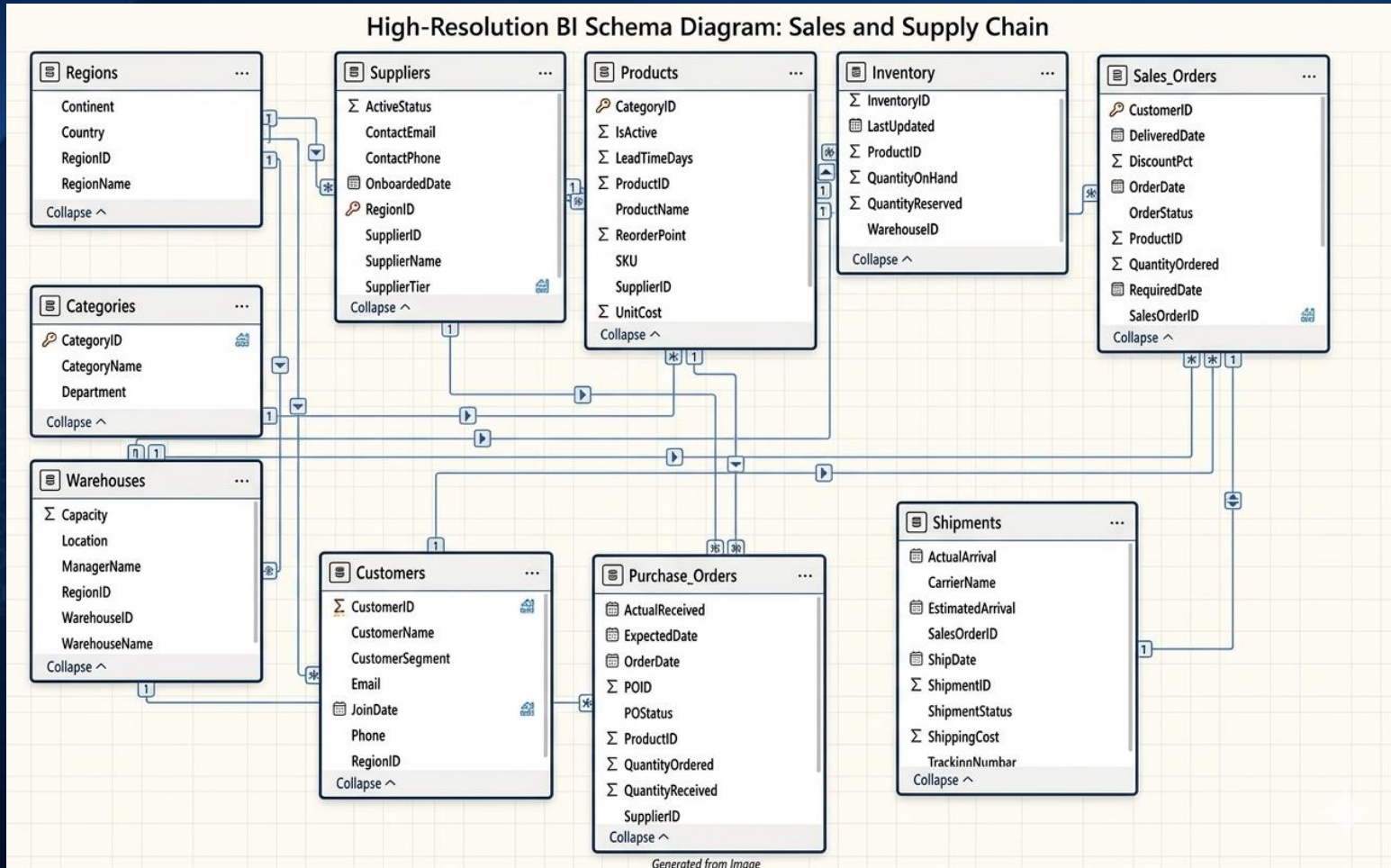
- Supplier delays
- Inventory imbalance
- Slow fulfillment
- Revenue visibility issue
- Customer profitability unclear
- Warehouse inefficiency
- Carrier performance issues
- Dead stock

These problems were directly impacting cost, revenue, and customer experience

DATA MODEL

Key Business Problems

Suppliers → Purchase Orders → Warehouses → Inventory → Customers → Sales → Shipments



This structure enabled complete end-to-end supply chain tracking

ANALYTICAL APPROACH

- Solved 8 core business problems
- Used SQL for data transformation & analysis

Techniques Used:

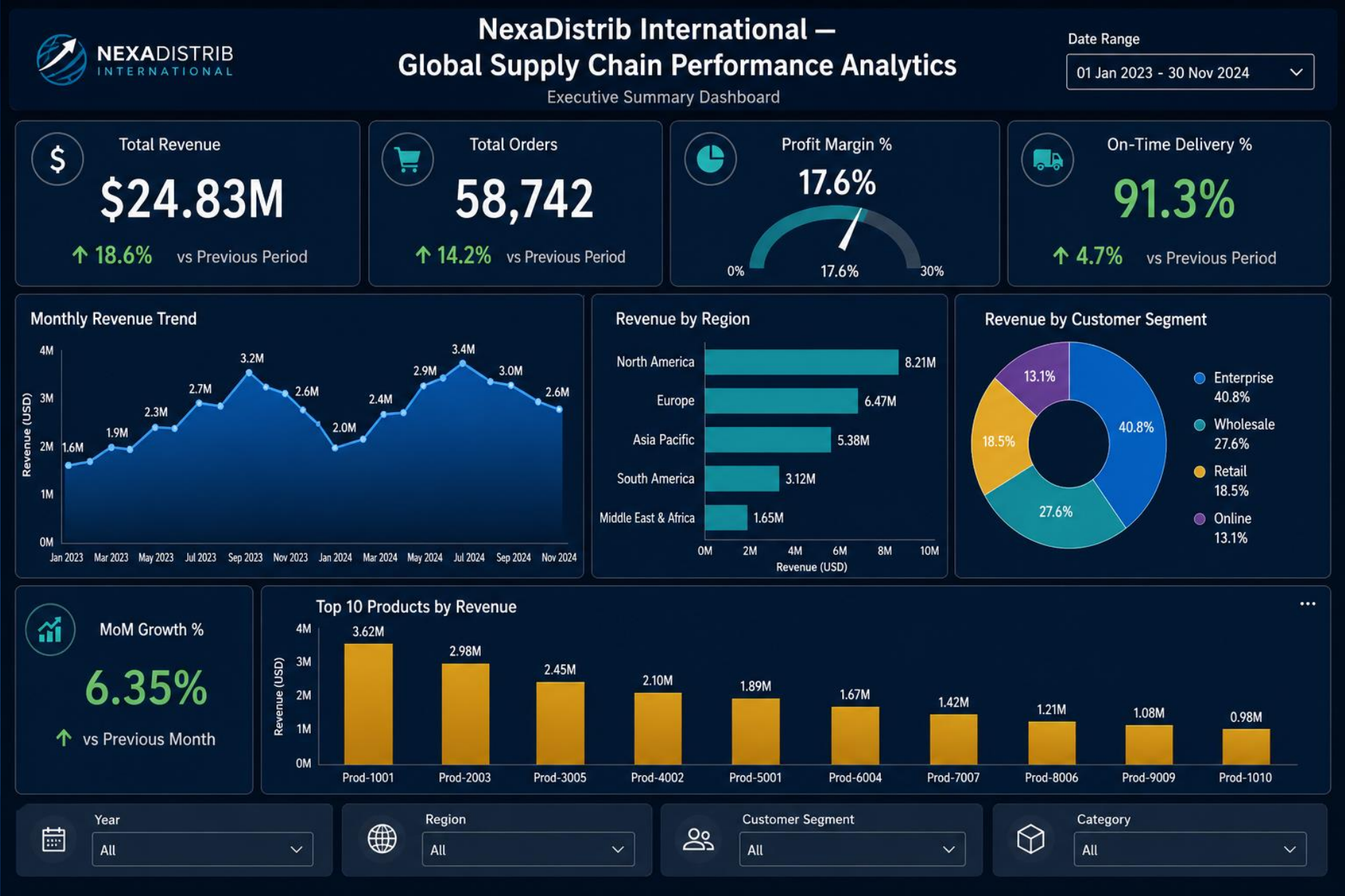
- CTEs (multi-step analysis)
- Window Functions (ranking, trends)
- Aggregations (KPIs)
- Date functions (time analysis)

```
SELECT
    SupplierID,
    COUNT(*) AS TotalOrders
FROM Purchase_Orders
GROUP BY SupplierID;
```

I used SQL techniques like aggregations and window functions to transform raw data into business insights. For example, this query calculates total orders per supplier, which helps evaluate supplier performance

DASHBOARD OVERVIEW

Executive Summary



DASHBOARD OVERVIEW

Inventory Analysis



NEXADISTRIB
INTERNATIONAL

Inventory & Warehouse Analysis

NexaDistrib International – Global Supply Chain Performance Analytics

01 Jan 2023 - 30 Nov 2024



Total Inventory Units

1,245,780

Units



Stockout Products

84

(8.2%)

Critical – Immediate Action Required



Low Stock Products

156

(15.1%)

Monitor & Replenish Soon



Warehouse Utilization %

72%

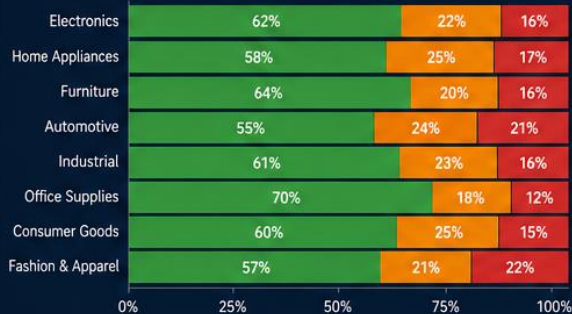
0%

50%

100%

Stock Status by Product Category

● Healthy ● Low Stock ● Out of Stock

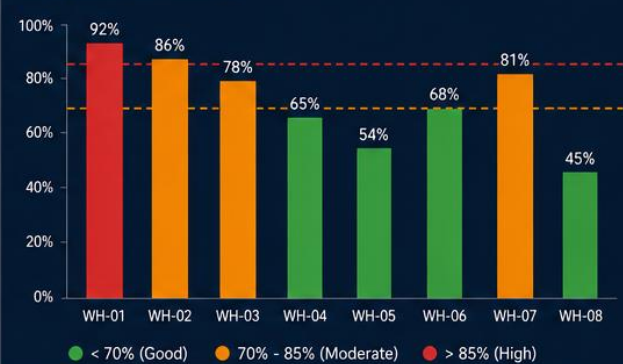


Product Inventory vs Reorder Point

ProductName	QuantityOnHand	ReorderPoint	Status
Wireless Headset	1,250	800	Healthy
Smart Watch	620	700	Low
LED Monitor 24"	0	500	Stockout
Bluetooth Speaker	1,150	900	Healthy
Gaming Keyboard	300	600	Low
Office Chair	2,150	1,500	Healthy
Desk Lamp	0	400	Stockout
USB-C Cable	800	600	Healthy
Hard Drive 1TB	450	700	Low
Air Conditioner	0	200	Stockout

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Warehouse Utilization %



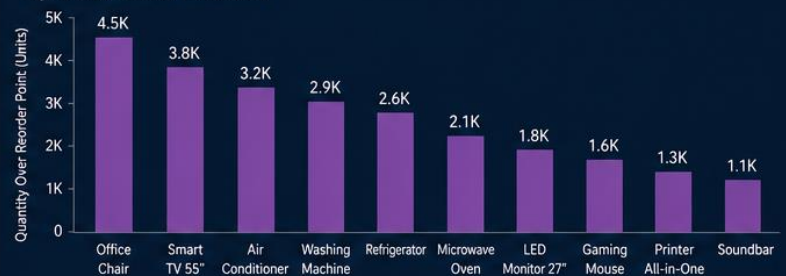
Inventory Distribution Across Warehouses

	WH-01	WH-02	WH-03	WH-04	WH-05	WH-06	WH-07	WH-08
Wireless Headset	1,200	950	780	1,050	650	890	1,100	620
Smart Watch	520	430	300	480	290	410	500	280
LED Monitor 24"	0	0	120	0	60	0	150	0
Bluetooth Speaker	800	650	500	780	420	600	700	450
Gaming Keyboard	200	180	150	220	120	160	190	110
Office Chair	1,100	950	800	1,250	700	900	1,150	650
Desk Lamp	0	40	30	0	20	0	10	0
USB-C Cable	600	480	420	550	380	500	580	350

Low Stock

High Stock

Top Overstocked Products



Warehouse

All



Category

All



Product

All



Region

All



DASHBOARD OVERVIEW

Supplier Performance



Supplier & Procurement Performance

NexaDistrib International – Global Supply Chain Performance Analytics

01 Jan 2023 - 30 Nov 2024



Total Purchase Orders

150

POs

Supplier On-Time %



Late Deliveries Count

36

↓ -12.2% vs Previous Period



Average Lead Time

6.4

Days

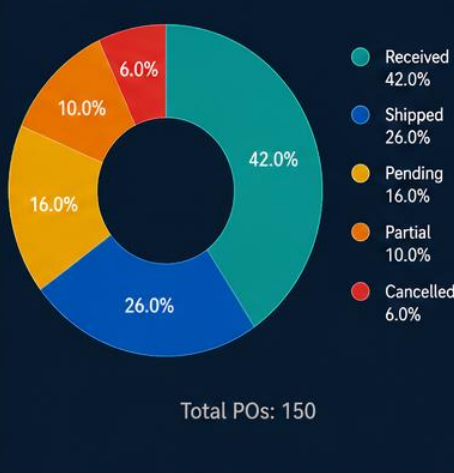
Supplier On-Time Delivery %



Supplier Performance Scorecard

SupplierName	Tier	Total POs	On-Time	Delayed	OnTime%
Apex Global Supplies	Gold	18	17	1	94.4%
Prime Materials Co.	Gold	15	14	1	93.3%
Global Trade Partners	Gold	12	11	1	91.7%
SupplyMax Ltd.	Gold	14	12	2	85.7%
Swift Procurement	Silver	13	11	2	84.6%
Reliable Sources Inc.	Silver	11	9	2	81.8%
Trust Supply Solutions	Silver	10	8	2	80.0%
Value First Suppliers	Silver	9	7	2	77.8%
Dynamic Goods Co.	Silver	8	6	2	75.0%
Nexus Components	Bronze	7	5	2	71.4%
Quality Plus Traders	Bronze	6	4	2	66.7%
Efficient Supplies	Bronze	6	4	2	66.7%
Metro Materials	Bronze	5	3	2	60.0%
Budget Sourcing	Bronze	5	2	3	40.0%
LowCost Distributors	Bronze	4	1	3	25.0%
Total		150	114	36	76.0%

PO Status Distribution



Purchase Orders by Supplier Tier



Lead Time Trend by Month



Supplier Tier



All



Supplier Name



All



Region



All



Product



All

DASHBOARD OVERVIEW

Sales & Delivery



Meaning Full Insights

1. Supplier Performance Insight

- Gold: 91% on-time
- Bronze: 58% on-time

Supplier reliability gap was a major operational risk.

2 . Inventory Insight

18% products low/out of stock

Inventory imbalance was causing lost sales and inefficiency.

3. Delivery Performance Insight

- Avg fulfillment: 6.2 days
- Some >10 days

Delivery inconsistency was impacting customer satisfaction

4. Revenue & Customer Insight

- Top 10% customers → 48% revenue
 - Enterprise segment highest
- Revenue concentration created a significant business risk.

Key Recommendations

- **Optimize Supplier Selection**
Prioritize high-performing suppliers and reduce dependency on unreliable vendors to improve delivery consistency
- **Implement Automated Reordering System**
Use data-driven reorder thresholds to ensure timely replenishment and prevent stockouts
- **Enhance Warehouse Utilization**
Redistribute inventory efficiently across warehouses to balance capacity and reduce operational costs
- **Improve Carrier Performance Management**
Select high-performing logistics partners and monitor delivery SLAs to reduce shipment delays
- **Reduce Dead Stock & Improve Inventory Turnover**
Identify slow-moving products and implement clearance or bundling strategies to free up working capital

These recommendations focus on improving efficiency across procurement, inventory, and logistics while directly impacting cost reduction and revenue growth

Expected Impact

- +20% Improvement in On-Time Delivery
- 50% Reduction in Stockouts
- 10-15% Increase in Revenue
- Significant Cost Reduction

These improvements can significantly enhance overall operational efficiency and profitability.

FINAL SUMMARY

End-to-End Supply Chain Analytics Solution

Covered complete workflow from procurement → inventory → sales → delivery

Advanced Data Analysis using SQL & Power BI

Applied CTEs, window functions, and DAX to transform raw data into actionable insights

Business-Centric Insights & Decision Support

Identified key inefficiencies in supplier performance, inventory planning, and delivery operations

Actionable Recommendations with Measurable Impact

Proposed solutions to improve efficiency, reduce costs, and increase revenue

This project showcases how I used data analytics to solve real-world supply chain challenges — from identifying operational inefficiencies to delivering actionable insights that drive better

THANK YOU

Any Questions?

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[GitHub](#)